

Environmental effects on meiotic recombination maps

A model linking meiotic chromatin to diversity and relative differentiation

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June 19, 2026

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Abstract

Environmental conditions can change both the number and the genomic positions of meiotic crossovers. Population-genetic models usually treat the recombination map as fixed, and experimental studies often combine these two changes. We separate them. In environment E , we write the map as $r_E(x) = R_E p_E(x)$, where R_E is total map length and $p_E(x)$ is the distribution of crossover positions. A local placement weight $q_E(x)$ determines $p_E(x)$. Meiotic chromatin accessibility contributes to this weight together with sequence context, chromosome structure, axis organisation, and crossover maturation. For two environments, the map change is $\Delta r = \Delta R \bar{p} + \bar{R} \Delta p$. The first term measures the change in total map length. The second measures the redistribution of crossovers along the chromosome and integrates to zero. A region gains relative crossover probability when its increase in placement weight exceeds its expected share of the genome-wide increase. The result also applies to regions with zero placement weight at baseline. We then model background selection, in which selection against deleterious alleles also removes linked neutral variants. In a local approximation, diversity rises where recombination rises. Relative differentiation falls in those regions when absolute divergence changes little. In the full model, diversity depends on recombination along the chromosome between neutral and selected sites, so the sign depends on changes across the surrounding map. These population-genetic effects accumulate over many generations. They can arise when the environment acts directly on meiosis in each generation or when an inherited state persists after exposure. The model specifies experiments and comparative tests that link meiotic chromatin, recombination-map change, and population-genomic variation.

1 Introduction

Crossovers are unevenly distributed along chromosomes. In budding yeast and plants, open chromatin, nucleosome organisation, histone modifications, DNA methylation, chromosome-axis proteins, sequence polymorphism, and structural context all contribute to their positions (Pan et al., 2011; Choi et al., 2013, 2018; Underwood et al., 2018; Fernandes et al., 2024; Son et al., 2025). Experiments that alter chromatin states also alter meiotic recombination, which supports a causal role for chromatin (Underwood et al., 2018; Fernandes et al., 2024; Son et al., 2025; Szymanska-Lejman et al., 2026).

The recombination map also responds to the environment. Temperature can change total crossover number, crossover positions, or both. The response varies among chromosomes, sexes,

genotypes, and temperature treatments (Lloyd et al., 2018; Modliszewski et al., 2018; De Storme and Geelen, 2020; Coulton et al., 2020). An analysis must distinguish total map length from the spatial distribution of crossovers.

Recombination also affects neutral genetic variation. Selection at one site can remove neutral variants at linked sites. Recombination weakens this association. Background selection and selective sweeps reduce more neutral diversity in regions with low recombination (Begun and Aquadro, 1992; Charlesworth et al., 1993; Hudson and Kaplan, 1995; Nordborg et al., 1996; Coop and Ralph, 2012). The same process can raise relative differentiation by reducing diversity within populations, even when absolute divergence remains near its background value (Charlesworth, 1998; Cruickshank and Hahn, 2014). Most models treat the recombination map as fixed. Theory on evolving maps shows that map history can change the strength and spatial distribution of background selection (Booker et al., 2022). Recombination maps can also differ between recently diverged ecotypes (Talbi et al., 2025).

We develop one model for these observations. We first separate total map length from the spatial distribution of crossovers. We then derive how an environmental change in local crossover placement alters that distribution. Next, we calculate the effects on diversity under local and nonlocal models of linked selection. We also derive the corresponding change in relative differentiation. Each equation states the conditions required for its use.

The map response and its population-genetic consequences can occur without epigenetic inheritance. The environment may act directly on cells entering meiosis, or the same environment may induce a similar state in each generation. Inheritance becomes a separate hypothesis when an altered state persists after the inducing environment has been removed.

2 From meiotic state to the recombination map

2.1 A model for crossover placement

Let $x \in [0, L]$ denote physical position along a chromosome and let E denote an environment or treatment. Define a nonnegative, integrable crossover-placement weight

$$q_E(x) = f(a_E(x), \mathbf{z}_E(x)), \quad q_E(x) \geq 0, \quad (1)$$

where $a_E(x)$ is meiotic chromatin accessibility. The vector $\mathbf{z}_E(x)$ contains other determinants of crossover placement, including chromatin marks, DNA methylation, sequence and structural variation, chromosome-axis organisation, synapsis, interference, sex, and the probability that a double-strand break matures into a crossover. The weight $q_E(x)$ measures local crossover propensity before normalisation. We leave the function f unspecified. The empirical question is whether accessibility explains part of q_E after the other determinants have been controlled.

Let

$$Q_E = \int_0^L q_E(x) dx, \quad p_E(x) = \frac{q_E(x)}{Q_E}, \quad \int_0^L p_E(x) dx = 1. \quad (2)$$

The density $p_E(x)$ gives the spatial distribution of crossover positions, conditional on a crossover occurring. Let R_E be total map length in environment E . The realised map is

$$r_E(x) = R_E p_E(x), \quad \int_0^L r_E(x) dx = R_E. \quad (3)$$

This factorisation allows total map length and spatial distribution to vary independently.

Scope. The model applies where meiotic chromatin explains a measurable part of crossover placement. A functional PRDM9 system may be present or absent. In PRDM9-directed systems, q_E can include motif binding and the chromatin changes recruited by PRDM9. In systems without functional PRDM9-directed placement, promoters, CpG islands, and other chromatin-associated features often explain more of the placement pattern (Baker et al., 2017; Talbi et al., 2025). The model should be tested by estimating the contribution of meiotic chromatin after accounting for the other terms in $\mathbf{z}_E$.

2.2 Total map length and spatial distribution

Compare a baseline environment $E = 0$ with a second environment $E = 1$. Write

$$\Delta r = r_1 - r_0, \quad \Delta R = R_1 - R_0, \quad \Delta p = p_1 - p_0,$$

and define the midpoints

$$\bar{R} = \frac{R_1 + R_0}{2}, \quad \bar{p} = \frac{p_1 + p_0}{2}.$$

Proposition 1 (exact total-shape decomposition). *For any two recombination maps of the form (3),*

$$\boxed{\Delta r(x) = \underbrace{\Delta R \bar{p}(x)}_{\text{total-map component}} + \underbrace{\bar{R} \Delta p(x)}_{\text{shape component}}.} \quad (4)$$

The two components satisfy

$$\int_0^L \Delta R \bar{p}(x) dx = \Delta R, \quad \int_0^L \bar{R} \Delta p(x) dx = 0. \quad (5)$$

Proof. Expanding the right-hand side gives

$$\frac{(R_1 - R_0)(p_1 + p_0) + (R_1 + R_0)(p_1 - p_0)}{2} = R_1 p_1 - R_0 p_0.$$

The integral statements follow from $\int p_0 = \int p_1 = 1$. $\square$

Explanation. The zero integral of the shape component follows from normalisation. It describes redistribution along the chromosome. Experiments must estimate ΔR separately to determine whether total recombination has changed.

We measure the amount of spatial redistribution in map units as

$$M_{\text{shape}} = \frac{\bar{R}}{2} \int_0^L |\Delta p(x)| dx. \quad (6)$$

The factor $1/2$ avoids counting the same redistributed map length twice. Let $M_{\text{total}} = |\Delta R|$. When either component is nonzero, define

$$\chi = \frac{M_{\text{shape}}}{M_{\text{shape}} + M_{\text{total}}}. \quad (7)$$

Values near one indicate a response dominated by spatial redistribution. Values near zero indicate a response dominated by total map-length change. We use χ as a descriptive summary of the observed response.

2.3 Change in local crossover placement

Let $q_1 = q_0 + \Delta q$ and $Q_1 = Q_0 + \Delta Q$, where $\Delta Q = \int_0^L \Delta q(x) dx$. Write $p = p_0$ for the baseline distribution.

Proposition 2 (exact redistribution of normalised placement). *For any admissible perturbation with $Q_1 > 0$,*

$$\Delta p(x) = \frac{\Delta q(x) - p(x)\Delta Q}{Q_1}. \quad (8)$$

Consequently,

$$\Delta p(x) > 0 \iff \Delta q(x) > p(x)\Delta Q. \quad (9)$$

Proof. Direct subtraction gives

$$\frac{q + \Delta q}{Q + \Delta Q} - \frac{q}{Q} = \frac{Q\Delta q - q\Delta Q}{Q(Q + \Delta Q)} = \frac{\Delta q - p\Delta Q}{Q + \Delta Q}.$$

The denominator is positive, so the numerator determines the sign. $\square$

Equation (8) remains defined when $q(x) = 0$. If an inactive region acquires positive placement weight, then $\Delta p(x) = \Delta q(x)/Q_1 > 0$. The equation also covers the appearance of new crossover-active regions.

Where $q(x) > 0$, define the fractional change $h(x) = \Delta q(x)/q(x)$ and its baseline placement-weighted mean

$$\langle h \rangle_p = \int_0^L p(x)h(x) dx = \frac{\Delta Q}{Q}. \quad (10)$$

Then Eq. (8) becomes

$$\Delta p(x) = p(x) \frac{h(x) - \langle h \rangle_p}{1 + \langle h \rangle_p}. \quad (11)$$

A region gains relative crossover probability when its placement weight increases by more than the placement-weighted genomic average. A uniform multiplicative change in q changes Q and leaves p unchanged.

Chromatin accessibility enters this result as one component of q_E . If accessibility changes while the other arguments of f remain fixed, then a small perturbation at locations with $a > 0$ gives

$$h(x) \approx \varepsilon_a(x)g_a(x), \quad \varepsilon_a(x) = \frac{\partial \log f}{\partial \log a}, \quad g_a(x) = \frac{\Delta a(x)}{a(x)}. \quad (12)$$

If $f(a) = a$, then $h = g_a$ exactly. More general functions allow accessibility to change double-strand-break initiation while other processes control crossover maturation. Models for q_E should include all available predictors of crossover placement.

3 Consequences for linked selection

3.1 Background selection across the recombination map

Background selection reduces neutral diversity when selection removes deleterious alleles and the neutral variants linked to them. Recombination separates neutral and selected sites, so the effect declines with genetic distance. Let $u(z)$ be the deleterious mutation-rate density at position z , and

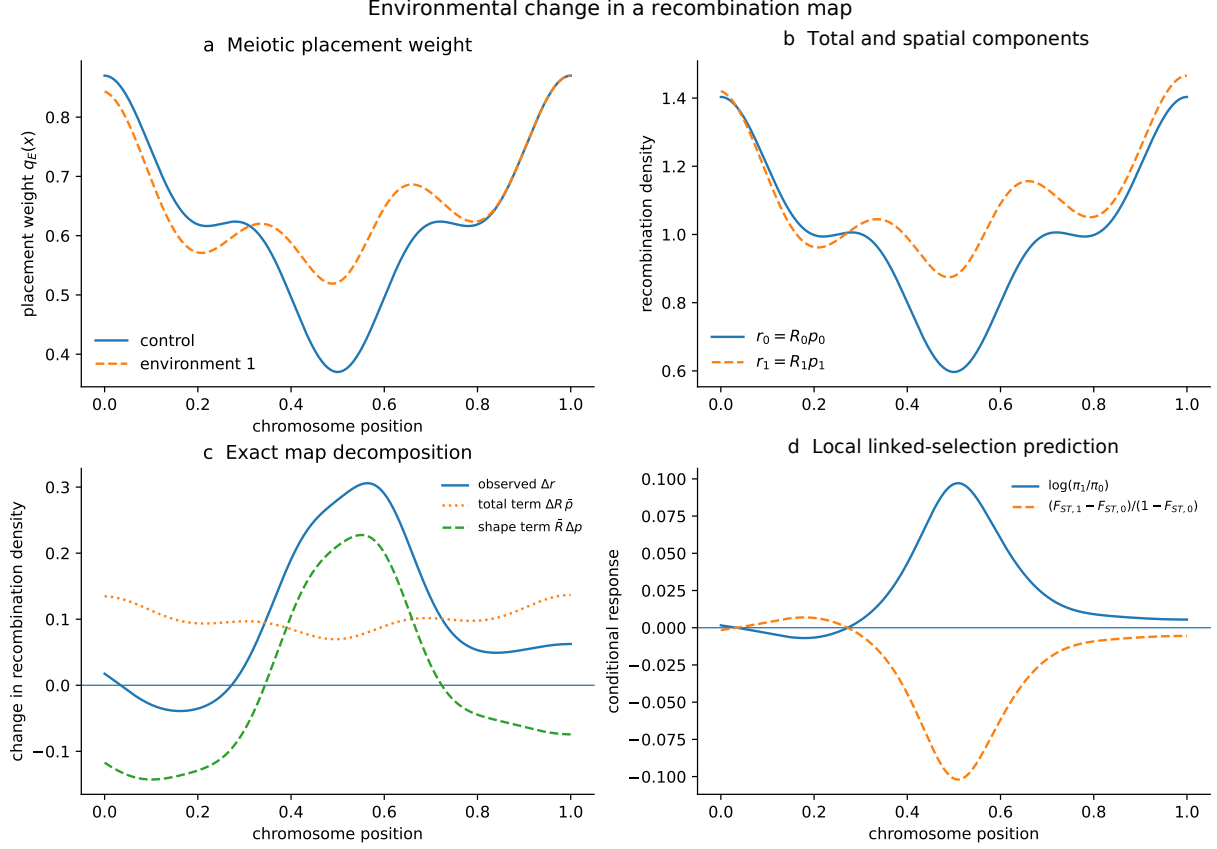


Figure 1: An example of an environmental change in the recombination map. **(a)** The environment changes the local crossover-placement weight $q_E(x)$, which includes meiotic chromatin and other determinants. **(b)** Total map length and the spatial distribution can change together. **(c)** Equation (4) separates the observed difference into a total component, which integrates to ΔR , and a shape component, which integrates to zero. **(d)** The local background-selection approximation predicts higher diversity where the local recombination rate rises. Relative differentiation falls in the same region when d_{xy} changes little. The nonlocal response in Eq. (16) averages map changes over the paths between neutral and selected sites.

let $s(z) > 0$ be the selection coefficient. Under map r , the genetic distance between positions x and z is

$$d_r(x, z) = \int_{\min(x, z)}^{\max(x, z)} r(y) dy. \quad (13)$$

We use the following standard approximation for the background-selection exponent:

$$E(x; r) = \int_0^L u(z) \frac{s(z)}{[s(z) + d_r(x, z)]^2} dz, \quad \pi(x; r) = \pi_{\text{neu}}(x) e^{-E(x; r)}. \quad (14)$$

Here $\pi_{\text{neu}}(x)$ is neutral diversity in the absence of linked selection. The contribution from a selected site decreases as its genetic distance from x increases.

For a small map perturbation δr , the change in genetic distance is

$$\delta d_r(x, z) = \int_{\min(x, z)}^{\max(x, z)} \delta r(y) dy. \quad (15)$$

Differentiating Eq. (14) gives

$$\delta \log \pi(x) = 2 \int_0^L u(z) \frac{s(z)}{[s(z) + d_r(x, z)]^3} \left[\int_{\min(x, z)}^{\max(x, z)} \delta r(y) dy \right] dz. \quad (16)$$

The ordered limits in Eq. (15) give the same positive genetic distance for selected sites on either side of x .

Equation (16) can also be written as

$$\delta \log \pi(x) = \int_0^L W_x(y) \delta r(y) dy, \quad W_x(y) \geq 0, \quad (17)$$

where $W_x(y)$ weights positions on the chromosomal segments between x and selected sites. Thus the change in diversity at x is a weighted average of map changes across those segments. Changes away from x can reinforce or offset the local change. A pointwise sign rule requires the local approximation below.

3.2 Local approximation

Suppose u , s , and r vary slowly over the linkage scale that contributes most to Eq. (14). Also suppose that the site of interest lies far enough from a chromosome boundary. Then

$$E(x; r) \approx \frac{\kappa u(x)}{r(x)}, \quad \kappa = 2 \quad (18)$$

for the kernel in Eq. (14) on a locally uniform, effectively infinite chromosome. Let $E_0(x) = \kappa u(x)/r_0(x)$. For a finite local map change,

$$\log \frac{\pi_1(x)}{\pi_0(x)} = E_0(x) \left[1 - \frac{r_0(x)}{r_1(x)} \right]. \quad (19)$$

Within this approximation, $r_1(x) > r_0(x)$ gives $\pi_1(x) > \pi_0(x)$.

For $q_0(x) > 0$, let $h = \Delta q/q_0$ and let $\eta_R = \Delta R/R_0$. From Eqs. (3) and (11),

$$\frac{r_1(x)}{r_0(x)} = (1 + \eta_R) \frac{1 + h(x)}{1 + \langle h \rangle_p}. \quad (20)$$

Substitution into Eq. (19) gives

$$\log \frac{\pi_1(x)}{\pi_0(x)} = E_0(x) \left[1 - \frac{1 + \langle h \rangle_p}{(1 + \eta_R)(1 + h(x))} \right]. \quad (21)$$

For small changes,

$$\frac{\Delta \pi(x)}{\pi(x)} \approx E_0(x) [\eta_R + h(x) - \langle h \rangle_p]. \quad (22)$$

When $\eta_R = 0$, the local diversity sign follows $h - \langle h \rangle_p$. When total map length changes, the total and spatial components may reinforce or oppose each other.

Both the local and nonlocal formulations give two general results. Regions with strong existing linked selection show larger fractional responses to a map change. Genome-wide diversity can also change when Δp integrates to zero, because the response is nonlinear and selected sites are unevenly distributed.

3.3 Timescale and map history

A recombination map can respond within one meiosis. Nucleotide diversity changes as genealogies turn over across many generations. If a map remains stable long enough, Eqs. (14)–(21) describe the resulting long-term values. If the map switches before genealogies relax, current diversity depends on the sequence of past maps (Booker et al., 2022). Forward or coalescent simulations can use that sequence directly. A one-generation experiment can estimate Δa , Δq , ΔR , and Δp . Tests of $\Delta\pi$ require recurrent exposure, a persistent state, or comparisons among populations with different long-term map histories.

4 Consequences for relative differentiation

For a Hudson-type sequence measure, write relative differentiation as

$$F_{ST}(x) = 1 - \frac{\pi_w(x)}{d_{xy}(x)}, \quad (23)$$

where π_w is diversity within populations and d_{xy} is mean pairwise divergence between populations. For two map regimes,

$$1 - F_{ST,1}(x) = [1 - F_{ST,0}(x)] \frac{\pi_{w,1}(x) d_{xy,0}(x)}{\pi_{w,0}(x) d_{xy,1}(x)}. \quad (24)$$

Its first-order form is

$$\Delta F_{ST}(x) \approx [1 - F_{ST}(x)] \left[\frac{\Delta d_{xy}(x)}{d_{xy}(x)} - \frac{\Delta \pi_w(x)}{\pi_w(x)} \right]. \quad (25)$$

Consider a map change that occurs after population separation, leaves the neutral mutation rate nearly unchanged, and acts mainly through within-population linked selection. In this regime, $d_{xy,1} \approx d_{xy,0}$. The local background-selection approximation then gives

$$F_{ST,1}(x) = 1 - [1 - F_{ST,0}(x)] \exp \left\{ E_0(x) \left[1 - \frac{r_0(x)}{r_1(x)} \right] \right\}. \quad (26)$$

Under these conditions,

$$\text{sign}(F_{ST,1} - F_{ST,0}) = -\text{sign}(r_1 - r_0). \quad (27)$$

When $\eta_R = 0$, the sign is opposite to $h - \langle h \rangle_p$. This sign rule applies to the stated local linked-selection regime with approximately stable d_{xy} .

The analysis should estimate d_{xy} . Ancestral polymorphism, mutation-rate variation, gene flow, introgression, ancestral linked selection, and structural variation can all change it. High F_{ST} together with low π_w and d_{xy} at or below its genomic background is consistent with linked selection in a low-recombination region. Elevated d_{xy} can arise from restricted gene flow or divergent selection, among other causes. The analysis should combine direct recombination maps, π_w , linkage disequilibrium, F_{ST} , d_{xy} , mutation-rate proxies, and demographic models.

Replicate populations exposed to similar environments may develop similar F_{ST} profiles even when selection acts on different loci. Evidence for the proposed mechanism requires a similar meiotic chromatin response, a similar Δp , and an F_{ST} response mediated mainly through π_w . Other possible causes include parallel selection, structural variants, and shared ancestral recombination maps.

5 Direct exposure, repeated induction, and inheritance

Three processes can produce recurrent differences in recombination maps.

Direct parental exposure. The environment reaches cells that undergo meiosis and changes q_E , R_E , or both. The altered gametes record a parental environmental effect.

Repeated environmental induction. The meiotic state resets each generation, and the same environment induces a similar map in each generation. Population-genomic effects can accumulate without persistence of the chromatin state between generations.

Inherited state. The inducing environment ends, and the altered meiotic state persists in one or more unexposed generations. A transgenerational inheritance claim requires persistence in the relevant unexposed generations.

A minimal model separates direct and inherited components. Let E_t be an environmental cue experienced by generation t , let $a_0(x)$ be the baseline meiotic accessibility profile, and write

$$a_t(x) = a_0(x) + \beta(x)E_t + m_t(x), \quad (28)$$

$$m_{t+1}(x) = \rho(x)m_t(x) + \gamma(x)E_t + \xi_t(x), \quad 0 \leq \rho(x) \leq 1. \quad (29)$$

The coefficient β gives the direct response within a generation. The state m_t is the transmitted component. The coefficient γ measures how exposure changes that component, and ρ measures its persistence after E_t returns to zero. Repeated induction can operate with $m_t = 0$. Evidence for inheritance requires $m_t \neq 0$ in the relevant unexposed descendants. The number of unexposed generations depends on the organism’s germline and life cycle.

We use the following terms. Repeated environmental induction is an environmental input to the recombination map. Persistent m_t is an inherited epigenetic input. We reserve the term feedback for systems in which the resulting genetic variation later changes the chromatin response or the interaction with the environment.

6 Tests of the model

The model contains a sequence of testable links. Evidence for population-genomic effects depends on support for the earlier meiotic and recombination links.

6.1 Estimate crossover number and position separately

The primary experiment should retain crossover data from individual meioses. Fit total crossover number or total map length by bivalent, meiocyte, sex, genotype, and treatment. This estimates R_E and the extent of crossover homeostasis. Then condition on the observed number of crossovers and model their positions along the chromosome. This estimates p_E and tests the placement model. A point-process model can include crossover interference. A multinomial or Dirichlet-multinomial model may be adequate for coarse genomic bins. Binwise correlations can mislead because p_E is compositional: an increase in one region produces relative decreases elsewhere. Conditional likelihoods or log-ratio methods account for this constraint.

Collect the chromatin data at the meiotic stage that produces the mapped gametes. Use somatic accessibility after establishing its relation to the meiotic state. Analyse male and female maps

separately when possible. Viability selection in gametes or zygotes can distort offspring-based crossover maps. Direct gamete assays and reciprocal designs can test for this bias.

Link	Quantity	Required design and expected result	Evidence against the proposed link
P0	Baseline coupling	In meiocytes, estimate whether accessibility and other chromatin variables explain crossover placement after controlling for sequence, structure, gene density, chromosome position, sex, and genotype.	Accessibility adds no predictive information to crossover placement in the study system.
P1	Environmental mechanism	Measure meiotic chromatin and crossover locations in matched control and treatment samples. Estimate Δq from the molecular predictors and test whether it predicts Δp through Eq. (8).	The observed Δp is repeatedly unrelated to the measured placement-state change after relevant covariates and meiotic timing are controlled.
P2	Total and spatial change	Estimate R_E and p_E separately, then report ΔR , M_{shape} , and χ .	A large ΔR with little Δp supports a change in total map length and gives little evidence for spatial redistribution.
P3	Long-run diversity	In replicated populations with recurrent exposure or a persistent map shift, test whether measured maps predict π and LD under forward or coalescent models that include demography and linked selection.	A fixed-map model fits as well as the measured map history, or the observed response differs from the prediction based on that history.
P4	Relative differentiation	Jointly model π_w , LD, F_{ST} , and d_{xy} . In the stated linked-selection regime, changes in F_{ST} should arise mainly through changes in π_w .	Mutation-rate variation, structural variants, gene-flow barriers, or selection at the loci under study explain the data better, and direct map data give no support for the proposed mechanism.
P5	Inheritance	Remove the cue and assay meiotic chromatin and recombination in the required unexposed descendants. Estimate persistence in Eqs. (28)–(29).	The state returns to baseline in all unexposed descendants. The result is consistent with direct or repeated induction and excludes persistent memory.

Table 1: Tests of the proposed mechanism. The zero integral of Δp follows from normalisation. Empirical tests concern changes in map shape, the contribution of meiotic chromatin to those changes, and the duration of the altered map.

6.2 Predict population-genomic effects

After estimating ΔR and Δp , use the measured map in forward or coalescent simulations. The model can include deleterious mutations, selective sweeps, demography, structural variants, and the frequency of environmental exposure. Simulations are needed because Eqs. (21) and (26) describe long-term local values. They should predict the spatial and temporal responses of π , LD, F_{ST} , and d_{xy} before comparisons with natural populations.

Comparative tests require ecological replication. Suitable designs include independent population pairs, replicate colonisations, and populations distributed along an environmental gradient. The analysis should include mutation rate, gene density, mappability, GC-biased gene conversion, centromeres, structural variants, sex-specific maps, and demographic history. LD-based recombination maps depend on demography and selection. Pedigree, gamete, and experimental maps provide more direct tests of the meiotic mechanism.

7 Biological settings and extensions

Climate and repeated stress. A sustained or recurring environment may induce a similar meiotic map in many generations. The immediate prediction is a reproducible change in R_E , p_E , or both. Population-genomic effects require enough generations for the map history to alter genealogies. Comparisons along thermal gradients should follow an experimental test of the treatment-to-map relation.

Colonisation and range expansion. A bottleneck reduces standing variation. A recombination-map shift changes the associations among the remaining alleles and the interference among selected loci. It may reduce Hill–Robertson interference in some regions or break favourable and locally adapted haplotypes in others. Its effect on adaptation depends on the positions of the recombination changes relative to selected loci and haplotypes.

Parallel ecological transitions. Replicate populations in similar environments may show similar meiotic-state and map responses. These responses may contribute to concordant patterns of diversity or relative differentiation. Shared ancestral maps, structural variants, genetic recombination modifiers, and parallel selection can produce similar patterns. Support for the environmental-chromatin mechanism requires direct and repeated observation of the meiotic-state and map changes.

Divergence with gene flow. Recombination affects linked selection and the persistence of multilocus associations under migration. An environmental map shift can change the genomic background used to infer barriers to gene flow. Peaks of F_{ST} have several possible causes. A map shift can also weaken or strengthen adaptive linkage, depending on its position. Joint models of migration, selection, structural variation, and map history are required.

8 Relation to existing theory and contribution

Separate parts of the model are established. Meiotic chromatin contributes to recombination placement. Temperature can alter crossover number and distribution. Linked selection relates diversity to recombination. Low within-population diversity can raise relative differentiation.

Changes in recombination maps can alter background-selection signatures (Underwood et al., 2018; Lloyd et al., 2018; Modliszewski et al., 2018; Cruickshank and Hahn, 2014; Booker et al., 2022).

Our contribution combines these results and makes four distinctions. First, Eq. (4) separates total map-length change from spatial change while allowing total recombination to vary. Second, Eq. (8) gives an exact sign rule for normalised placement, including regions with zero baseline weight. Third, Eqs. (16) and (21) separate the full nonlocal response from its local approximation. Fourth, Eqs. (28)–(29) distinguish direct exposure, repeated induction, and inheritance. These distinctions define an experimental sequence in which each link can be tested separately.

Booker et al. (Booker et al., 2022) model background selection under changing recombination rates. The present model adds an environmentally responsive meiotic input, the total-shape decomposition, and linked mechanistic and comparative tests. It predicts changes in population-genomic variation when an environmental cue alters recombination and the altered map recurs or persists over enough generations.

9 Assumptions and limits

Chromatin accessibility. Accessibility may facilitate double-strand-break initiation. Crossover outcome also depends on axis organisation, synapsis, sequence divergence, structural heterozygosity, interference, and repair-pathway choice. The variable q_E includes these contributions. Experiments can estimate it from crossover data and the available molecular predictors.

Double-strand breaks and crossovers. A perturbation can increase break formation while leaving crossover number unchanged. It can also redirect repair into noncrossovers or change crossover maturation. Crossover maps provide the primary response variable. Break maps help identify the stage at which the perturbation acts.

Crossover homeostasis. The obligate crossover and interference constrain crossover number. Temperature experiments also show changes in R_E . Normalisation makes the shape component in Eq. (4) integrate to zero. The total observed map can still change.

Local and nonlocal diversity responses. Equation (21) gives a pointwise sign under the local background-selection approximation. Equation (16) gives the full response. The distribution of selected sites, chromosome boundaries, structural variants, and changes in flanking regions can alter diversity at the site of interest.

Absolute divergence. The approximation $\Delta d_{xy} \approx 0$ defines one regime. The analysis should test it. Ancestral selection, mutation-rate variation, introgression, and long-running map differences can change d_{xy} .

Recombination and adaptation. Higher recombination can separate favourable alleles from deleterious backgrounds and reduce interference. It can also break coadapted or locally adapted haplotypes. The model predicts a change in linkage structure. Genetic architecture and the positions of selected loci determine the effect on adaptation.

Evidence for inheritance. Classify offspring effects according to which generations and germline precursors received direct exposure. A test of transgenerational inheritance requires the appropriate unexposed generations. The map-plasticity model still applies when the chromatin state resets in every generation.

10 Conclusion

Studies of environmental effects on recombination should estimate total map length and the spatial distribution of crossovers separately. The factorisation $r_E = R_E p_E$ and the decomposition $\Delta r = \Delta R \bar{p} + \bar{R} \Delta p$ provide this separation. A measured change in crossover-placement weight gives an exact rule for the change in p_E . Linked-selection theory then gives conditional predictions for diversity and relative differentiation.

Tests should follow the same sequence as the model: measure meiotic state, estimate crossover number and positions, establish the frequency or persistence of the map change, and compare the resulting population-genomic patterns with simulations. The formulation states the conditions under which an environmental effect on meiotic chromatin can leave a population-genomic signal.

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